

- 2 Human immunodeficiency virus (HIV) is an example of a virus that spreads from other animals to humans where it causes disease.

Fig. 2.1 shows a chimpanzee, *Pan troglodytes troglodytes*. Chimpanzees can carry the simian immunodeficiency virus (SIV) which is similar to HIV.

It is thought that chimpanzees who carry antibodies for SIV do not become ill if infected with HIV.

This has been investigated by scientists developing potential vaccines for HIV.



Fig. 2.1

Tests were carried out to see if antibodies against SIV present in chimpanzees bind to HIV antigens.

Test strips were prepared which contained several different HIV antigens.

When samples are applied to the test strip a line will appear in the control region. If the sample contains antibodies to the HIV antigens present on the strip, additional lines will also appear.

Samples of chimpanzee faeces were collected from a number of sites in Gabon in central Africa.

The faecal samples were prepared and then applied to the test strips.

- (a) (i) Suggest why the investigators collected faecal samples from chimpanzees rather than plasma samples.

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- (ii) Suggest a method for preparing the faecal samples before applying them to the test strips.

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- (iii) The scientists collected 608 faecal samples from 224 individual chimpanzees.

Suggest a laboratory method that could be used to identify whether two faecal samples belong to the same individual or not.

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Samples from humans and chimpanzees were applied to the test strips.

1. Plasma samples from humans who are infected with HIV (HIV+)
2. Plasma samples from humans who are not infected with HIV (HIV-)
3. Faecal samples from chimpanzees who are infected with SIV (SIV+)
4. Faecal samples from chimpanzees who are not infected with SIV (SIV-)

The results are shown in Fig. 2.2.

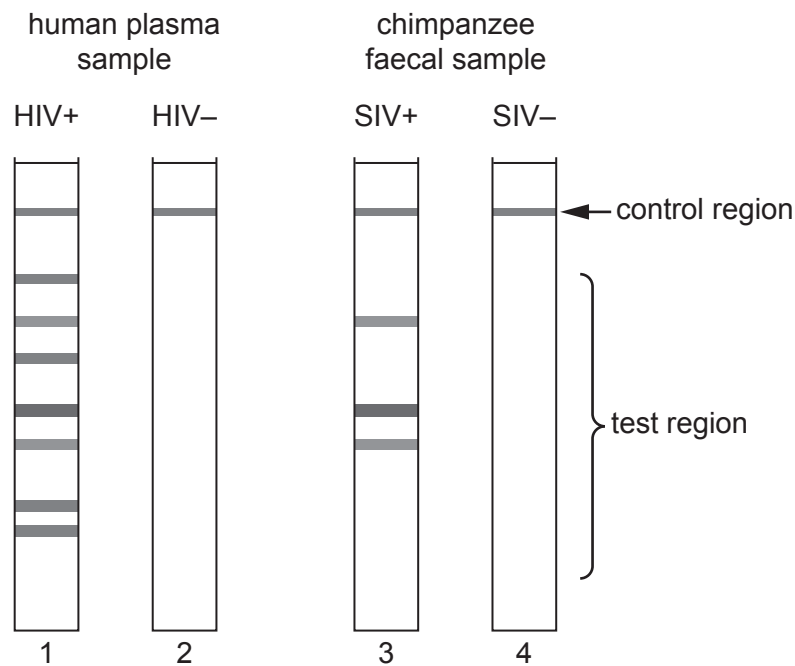


Fig. 2.2

(b) Suggest conclusions that can be drawn from the results shown in Fig. 2.2.

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Question 2 continues on page 12.

(c) HIV infection may lead to HIV/AIDS which, if left untreated, may cause death. The effects of SIV infection in chimpanzees are usually less severe.

A group of scientists investigated the effect of SIV infection on the life expectancy of a population of chimpanzees living in the wild.

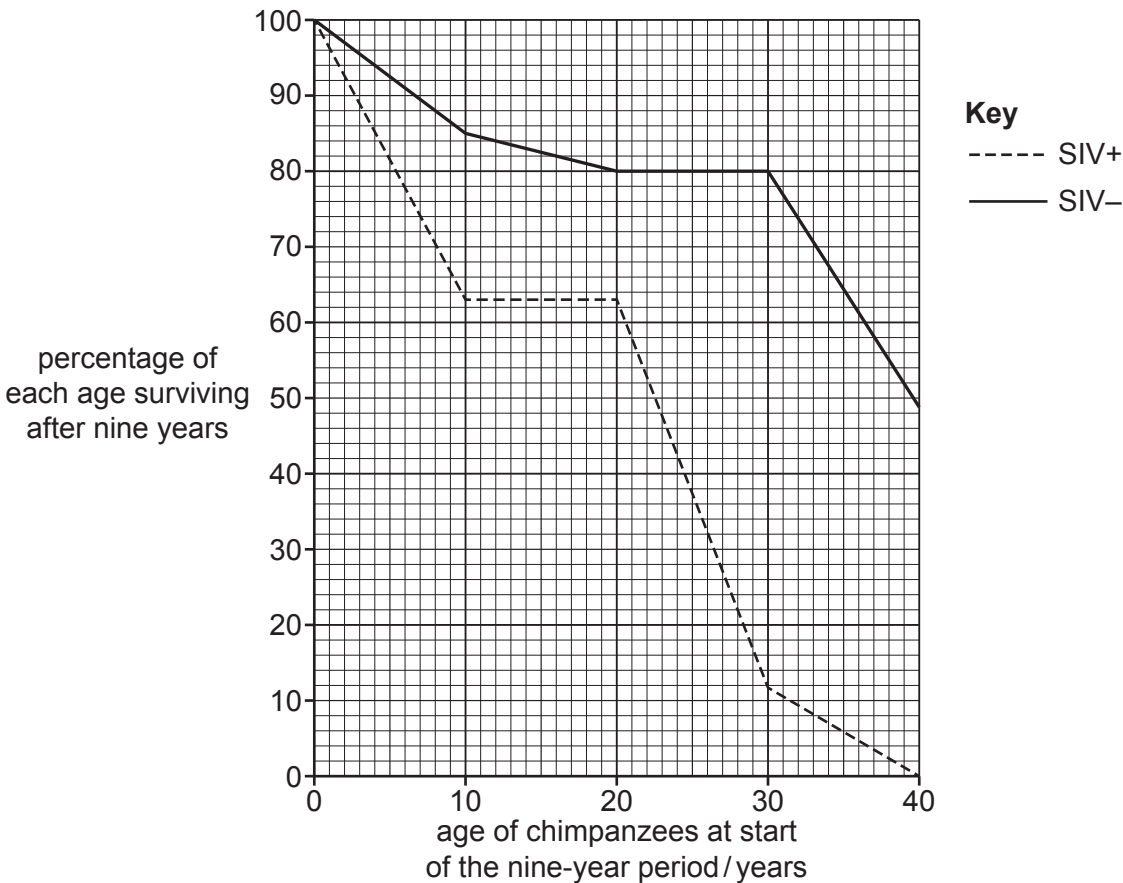
- The population of 94 chimpanzees was observed for a nine-year period.
- The ages of all the chimpanzees in the population were estimated.
- The chimpanzees were observed each day.
- The numbers of dead and absent chimpanzees were recorded each day.
- Faecal samples of all the chimpanzees in the population were tested for SIV antibodies.

The results of the investigation are summarised in Table 2.1 and Fig. 2.3.

Table 2.1

SIV status	original number of chimpanzees in population	number of chimpanzees who died or disappeared during the study
SIV–	77	11
SIV+	17	7

The percentage of chimpanzees of each age remaining in the population are represented in Fig. 2.3.



- (i) Fig. 2.3 presents the results of the investigation as the percentage of chimpanzees surviving at each age.

Explain the advantages of presenting the results of this study as percentages.

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The investigators concluded that chimpanzees infected with SIV die at an earlier age than uninfected chimpanzees.

The investigators decided to test their conclusion statistically.

- (ii) State a null hypothesis the scientists could make before carrying out their statistical test.

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- (iii) The investigators carried out their statistical analysis and found that they could reject the null hypothesis at $p < 0.05$.

Explain what is meant by the term $p < 0.05$.

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- (iv) Suggest why the data presented in Fig. 2.3 may not be an accurate representation of the effect of SIV infection on the life expectancy of the chimpanzees.

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[Total: 14]